آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-I Physics Annual Examinations 2024

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part I Physics. It contains comments on candidates' responses to the 2024 HSSC-I Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

Most candidates achieved success in constructing good responses, especially in the questions related to the concepts of uncertainty in the measurement, projectile motion, rotational motion, fluid dynamics, work-energy theorem and thermodynamics. However, some of the candidates struggled with questions, based on vectors, oscillation, absolute energy and word problems. Nonetheless, it is essential for teachers to concentrate on the following content and provide candidates with more drills and practice to foster a solid understanding.

- Recognise the importance of unit conversion and extracting data from the question.
- Identify the demand of the question and application of correct formula with correct substitution and calculations.
- Understand the key words and command words used in the question.

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS

Constructed Response Questions (CRQs)

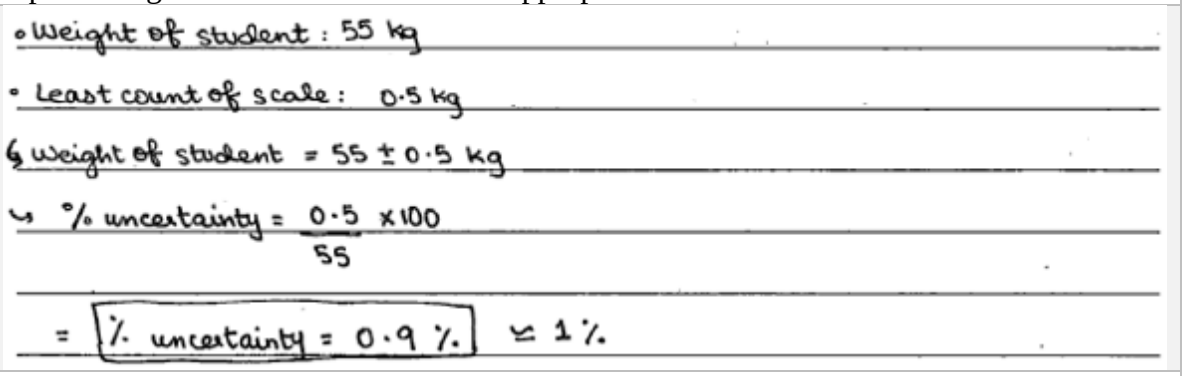
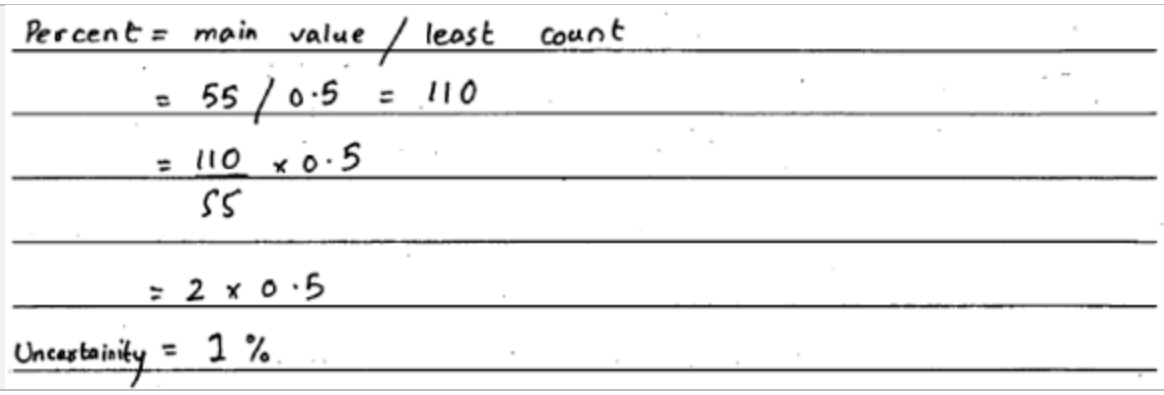
Question No. 1	
Question Text	The weight of a student on a certain weighing scale measures about 55 kg. If the least count of the machine is 0.5 kg, then calculate the percentage uncertainty in the measurement.
SLO No.	1.3.2
SLO Text	Solve word problems related to the uncertainty in the derived quantity;
Max Marks	03
Cognitive Level	A*
Checking Hints	1 mark for writing the correct formula of uncertainty 1 mark for substituting the correct values in the formula 1 mark for calculating the correct percentage uncertainty
Overall Performance	The overall performance of the cohort on this question was highly commendable. The majority of candidates demonstrated a clear understanding of the question's requirements, showcasing their understanding of absolute and percentage uncertainty. This level of competence indicates a solid grasp of the concepts and their implications, showcasing candidates' proficiency in the subject matter. Moreover, their ability to apply these principles to different scenarios highlights their analytical skills.
Description of Better Responses	*Better responses* were observed, reflecting candidates' proficiency in extracting relevant data from the question and utilising the correct formula for calculating percentage uncertainty in the measurement of weight. $$\%\ uncertaincy = \frac{absolute\ uncertainty}{measurement} * 100$$ Moreover, these candidates correctly recognised that the given least count of an instrument represents absolute uncertainty, and they applied this understanding to the scenario, representing their measurements in the appropriate standard form.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, most of the candidates struggled to differentiate between absolute and relative uncertainty. They used incorrect formulas in terms of placing the denominator into the numerator and vice versa and made incorrect substitutions, which resulted in inaccurate answers.

Image of Weaker Response	

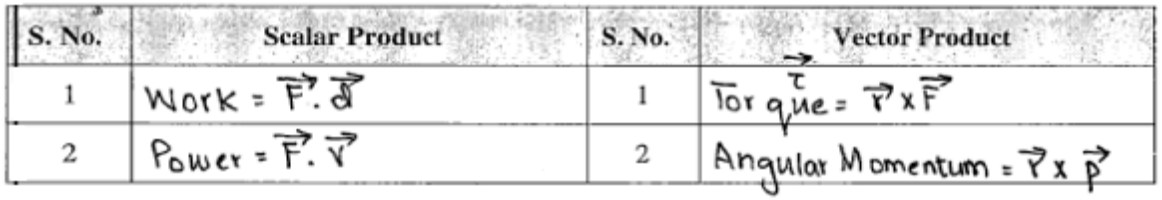
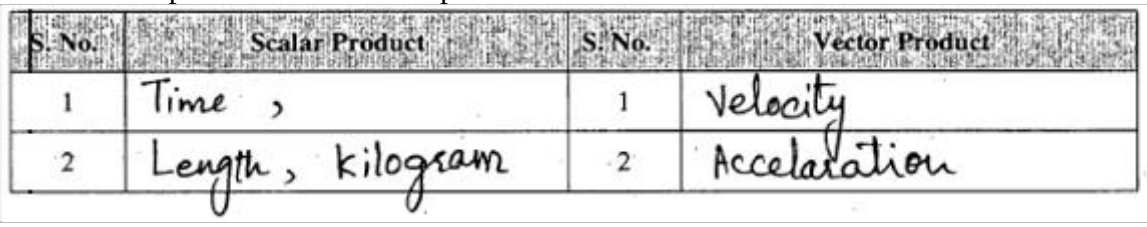
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy** Used for this SLO**	**Assessment Strategies**
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

- Subject teachers are encouraged to discuss the concepts of uncertainty in the measurement and explain the choice of instrument, with different least counts, can affect the uncertainty.
- An activity can be performed to understand the importance of least count by measuring the length of the same object with different instruments like a ruler, vernier calliper and screw gauge. By comparing the measurements, students can calculate uncertainty and able to comment about the better choice of the instrument.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

<table>
<tr><th colspan="2">Question No. 2</th></tr>
<tr><td>Question Text</td><td>Write any TWO physical quantities that exemplify each of the following.
<table>
<tr><th>S. No.</th><th>Scalar Product</th><th>S. No.</th><th>Vector Product</th></tr>
<tr><td>1</td><td></td><td>1</td><td></td></tr>
<tr><td>2</td><td></td><td>2</td><td></td></tr>
</table></td></tr>
<tr><td>SLO No.</td><td>2.4.2</td></tr>
<tr><td>SLO Text</td><td>Exemplify the scalar product of two vectors in terms of angle between them.</td></tr>
<tr><td>Max Marks</td><td>02</td></tr>
<tr><td>Cognitive Level</td><td>U</td></tr>
<tr><td>Checking Hints</td><td>1 mark for writing TWO examples of the scalar product
1 mark for writing TWO examples of the vector product</td></tr>
<tr><td>Overall Performance</td><td>The overall performance on this question was positive. However, some candidates struggled to understand the question’s requirements, resulting in less effective responses. Many candidates missed key details and, instead of providing examples of dot and cross products, incorrectly gave examples of scalar and vector quantities.</td></tr>
<tr><td>Description of Better Responses</td><td>In better responses, candidates effectively addressed the question’s requirements. Their responses demonstrated a clear understanding of dot and cross products, as they not only provided examples of physical quantities defined by these operations but also correctly stated the associated formulas.</td></tr>
<tr><td>Images of Better Responses</td><td> 
</td></tr>
<tr><td>Description of Weaker Responses</td><td>In weaker responses, candidates misunderstood the question’s requirements or did not read it carefully, resulting in incorrect examples. Their responses focused solely on the terms scalar and vector, leading them to provide examples of scalars and vectors instead of the correct examples of dot and cross products.</td></tr>
<tr><td>Images of Weaker Responses</td><td> 
</td></tr>
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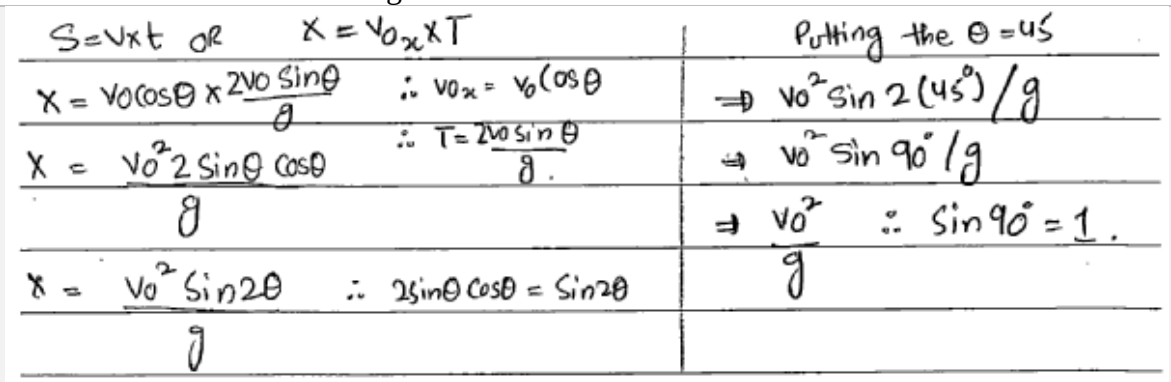
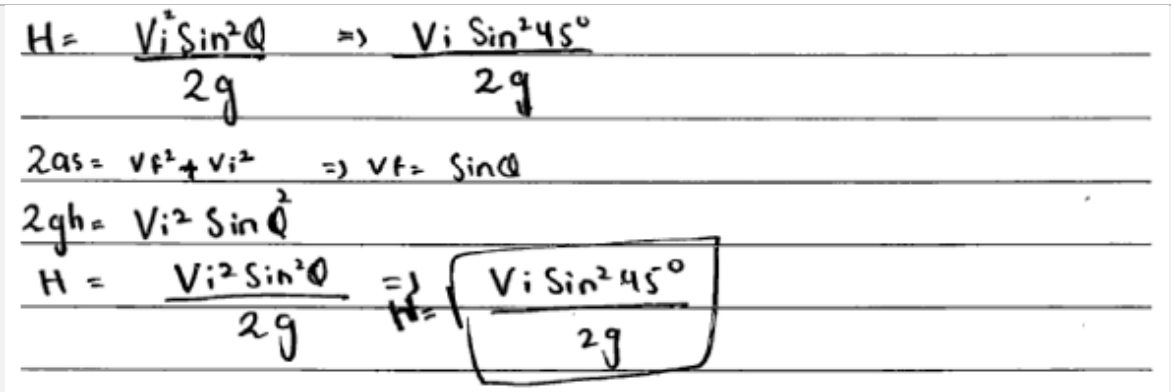
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Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:

- Teachers are encouraged to create practice questions with varied requirements to ensure that students read the questions carefully. Emphasising command words and providing complete, accurate answers will help students improve their attention to detail.
- Subject teachers must practice their students to read the question carefully and identify the demand of the question.

Question No. 3	
Question Text	During a practice session of long jump, a coach trains an athlete to take off the jump at an angle of 45°. Derive a mathematical expression for the maximum range of the athlete.
SLO No.	3.6.3
SLO Text	Derive the relation for a. time of flight, b. maximum height, c. horizontal range of a projectile.
Max Marks	03
Cognitive Level	A
Checking Hints	1 mark for writing each highlighted mathematical statement/ equation (3 required)
Overall Performance	The overall performance on this question was commendable. Candidates showed a strong ability to identify and analyse the given situation. Additionally, most recognised that the optimal angle of projection for achieving maximum range is 45°, demonstrating a solid understanding of projectile motion principles and effective application of their knowledge. Their proficiency in pinpointing key factors in projectile motion highlights their strong analytical skills.

Description of Better Responses	*Better responses* reflected candidates' proficiency in gathering the correct components needed to establish a mathematical relationship for maximum range. They effectively used the projectile's horizontal motion to derive the formula for range. By applying their understanding of how to maximise this relationship, they successfully developed the correct formula for the maximum range.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates were misled by the word 'maximum' and ended up deriving the formula for maximum height instead of maximum range. Some candidates correctly identified the requirement of the question but mistakenly employed the formula for the time to reach the maximum height instead of the correct formula for the total time of flight, leading to errors in their derivation. Other responses correctly established the relationship for range but struggled to apply the condition for maximum range in their calculations.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

- It is advised to the subject teachers that the motion of the projectile can be demonstrated and discussed its horizontal and vertical motion separately, giving clear knowledge about the application of gravitation force and its role in the motion of a projectile.
- Teachers may connect the concept of projectile motion to practical applications and compare their motion with the motion in the absence of air resistance.
- Subject teachers may engage their students in activities like launching projectiles at different angles and velocities.
- Teachers are encouraged to continue practicing and applying concepts in different contexts will further enhance their mathematical approach towards problem-solving abilities and ensure continued success in similar scenarios.

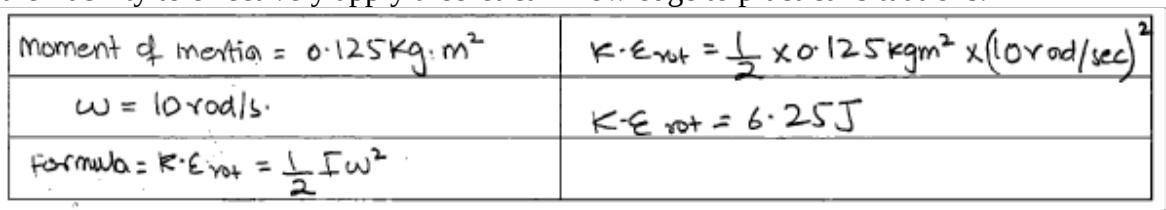
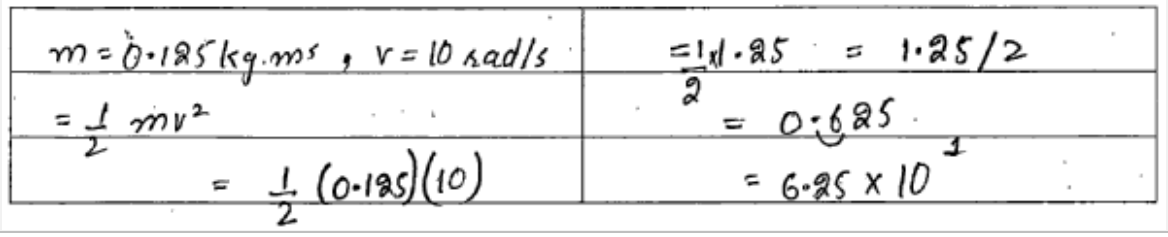
Question No. 4	
Question Text	A solid disc with 0.125 kg.m^2 of moment of inertia is rotating about its central axis with an angular velocity of 10 rad/s. Calculate the rotational kinetic energy of the disc.
SLO No.	5.5.2
SLO Text	Derive an expression for rotational kinetic energy and use this expression for solving word problems.
Max Marks	02
Cognitive Level	A
Checking Hints	1 mark for using the correct formula/ substituting values 1 mark for the correct answer
Overall Performance	The overall performance on this question was commendable. Most of the candidates performed well in calculating the rotational kinetic energy of the solid disc. They demonstrated a clear understanding of the question and applied the correct methods to arrive at accurate answers. Their responses were clear, reflecting a strong grasp of the topic and effective problem-solving skills. However, some candidates focused solely on kinetic energy and instead of adapting the data to rotational motion, incorrectly identified the moment of inertia as mass and rotational velocity as linear velocity, leading to calculations for linear kinetic energy rather than the required rotational kinetic energy.
Description of Better Responses	In *better responses*, candidates correctly identified and used the moment of inertia and angular velocity for a solid disc to calculate the rotational kinetic energy. These well-executed calculations demonstrate their solid understanding of mathematical concepts and their ability to effectively apply theoretical knowledge to practical situations.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates faced challenges in correctly writing the formula for the rotational kinetic energy of a solid disc and mixed up the formula of rotational kinetic energy and linear kinetic energy. Moreover, some candidates overcomplicated the moment of inertia of the solid disc and ignored crucial information and established relation $K.E = \frac{1}{4}mr^2\omega^2$ which was not needed in the given scenario.

Image of Weaker Response	

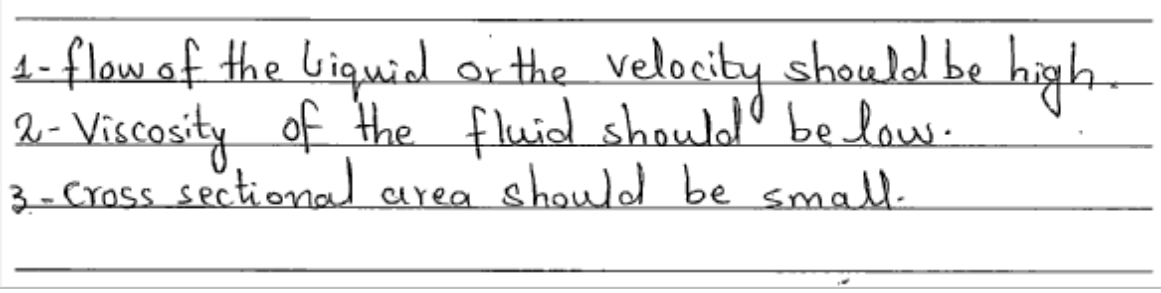
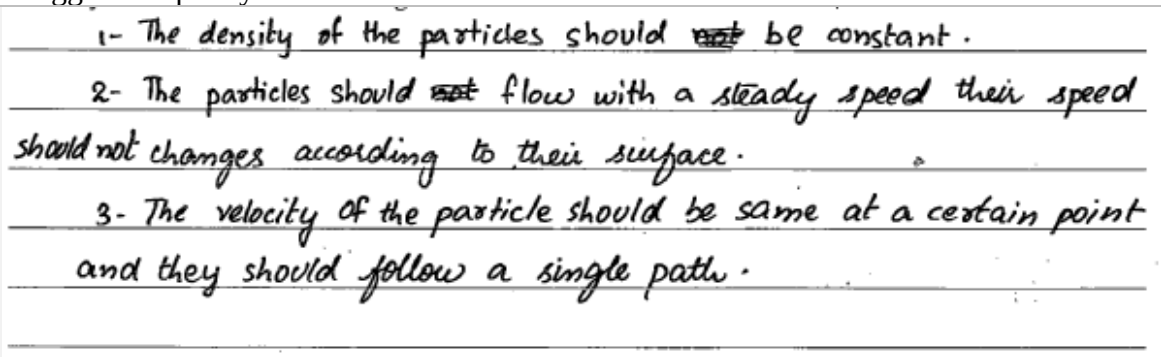
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Any Additional Suggestion:

Teachers are advised to practice their students to focus on reviewing and understanding the appropriate formulas and given conditions in the question by providing them with different worksheets and assignments.

Question No. 5	
Question Text	State any THREE conditions for the generation of turbulent flow.
SLO No.	6.1.2
SLO Text	State the conditions required for turbulent flow.
Max Marks	03
Cognitive Level	K
Checking Hints	1 mark for writing each condition (Any 3 required)
Overall Performance	The cohort performed exceptionally well on this question, demonstrating a strong ability to identify the requirements of the question. Most of the candidates precisely wrote the conditions. Such well-attempted responses indicate a solid grasp of relevant concepts.

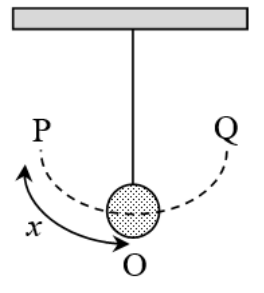
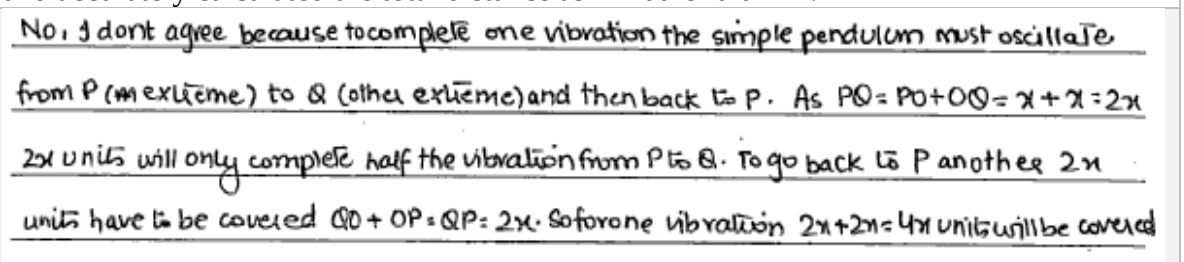
Description of Better Responses	The cohort accurately distinguished between laminar and turbulent flow, demonstrating precision in their answers. Some of the conditions written by candidates are as follows. 1. Lower fluid viscosity: Fluids with lower viscosity are more prone to transitioning to turbulent flow. 2. Presence of irregularities: Rough or irregular surfaces can cause turbulence in the flow. 3. Disruptions or obstacles: Disturbances in the flow path, such as obstacles or abrupt changes, can trigger turbulence by disturbing the smooth flow of the fluid. 4. High flow rates: Turbulence is more likely to occur in situations with high flow rates. 5. High Reynold number: A High Reynold number typically over 4000 signifies the turbulent flow of fluid in a pipe.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* revealed instances of incorrect identification, some responses demonstrated a deviation from the specified command words, resulting in inaccuracies in writing conditions. Some candidates wrote about the conditions for laminar flow, while others discussed both laminar and turbulent flow, detailing their differences. However, they struggled to specify the conditions that characterise turbulent flow.
Image of Weaker Response	

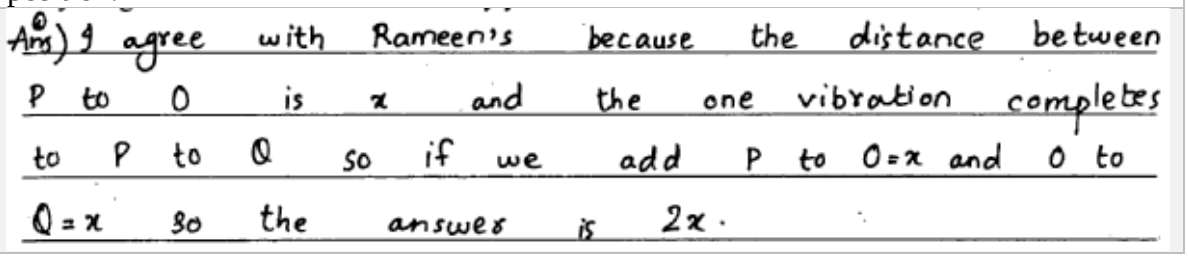
Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

- Teachers can play a crucial role in helping their students to enhance their focus on the requirement of the question, explain the importance of understanding what the question is asking before attempting an answer.
- Encourage candidates to make an experimental setup to demonstrate different conditions that cause turbulent in the flow of fluids.
- Encouraging candidates to closely examine the content asked in the question and consider their distinguishing attributes can prevent such errors.

Question No. 6	
Question Text	Rameen observes a simple pendulum oscillating between P and Q, over the mean position O as shown in the given diagram.  She claims that in one complete vibration, the distance covered by the bob is 2x units. Do you agree with Rameen's claim? Justify your stance with a reason.
SLO No.	7.5.1
SLO Text	Show the motion of a simple pendulum is SHM.
Max Marks	02
Cognitive Level	U
Checking Hints	1 mark for writing the stance 1 mark for writing the correct reason
Overall Performance	The candidates' overall performance on this question was generally satisfactory, demonstrating a sound grasp of the key features of oscillation. However, some responses lacked clarity in their reasoning, offered weak justifications, misunderstood the diagram and made wrong assumptions.
Description of Better Responses	*Better responses* demonstrated strong knowledge of vibrational motion, noting that vibration is always divided into four segments. Candidates effectively identified the amplitude of vibration from the provided diagram and paid close attention to the command words. They not only challenged the claim but also provided well-reasoned justifications that one complete vibration, the distance covered by the bob is 2*x* units for their position and accurately calculated the total distance as 4x rather than 2*x*.
Image of Better Response	 No, I dont agree because to complete one vibration the simple pendulum must oscillate from P (an extreme) to Q (other extreme) and then back to P. As PQ = PO + OQ = x + x = 2x 2x units will only complete half the vibration from P to Q. To go back to P another 2x units have to be covered QO + OP = QP = 2x. So for one vibration 2x + 2x = 4x units will be covered

Description of Weaker Responses	Candidates' responses to this question varied, with some agreeing but offering illogical explanations related to the vibration, often misled by the diagram and mistakenly treating half of a vibration as a full one. Other candidates confused distance with displacement and incorrectly justified that the distance is zero because the pendulum bob returns to its original position.
Image of Weaker Response	 Ans) I agree with Rameen's because the distance between P to O is x and the one vibration completes to P to Q so if we add P to O=x and O to Q=x so the answer is 2x.

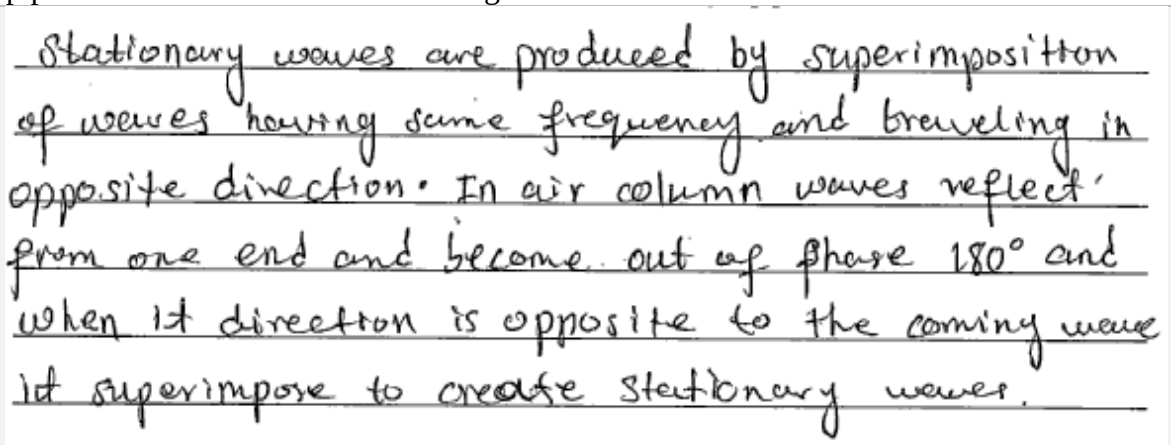
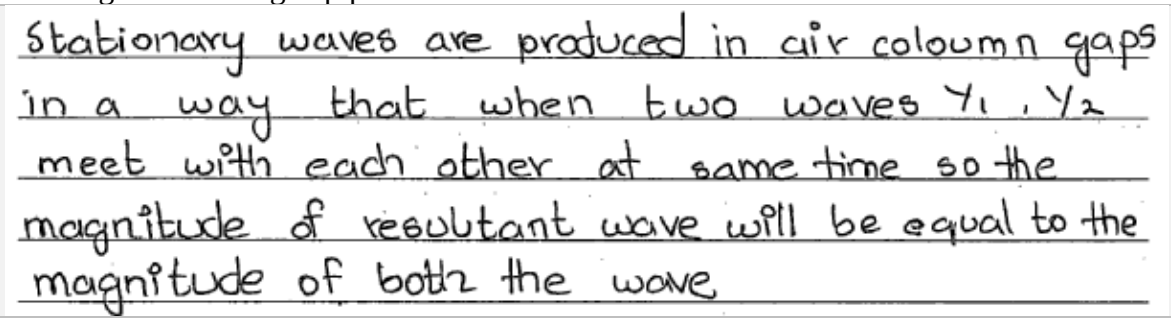
Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

Subject teachers are suggested to make the understanding of their students, the correct application of relevant concepts used in these types of questions.

Question No. 7	
Question Text	Explain how stationary waves are produced in an air column organ pipes.
SLO No.	8.4.5
SLO Text	Identify the formation of stationary waves in a vibrating air column.
Max Marks	03
Cognitive Level	U
Checking Hints	1 mark for each point (3 required)
Overall Performance	The overall performance of the cohort on this question was positive. Candidates showed a solid understanding of standing waves, with most providing accurate explanations of how standing waves are produced in an air column. However, some candidates struggled to grasp the specific requirements and instead wrote formulas related to fundamental frequency and harmonics.

Description of Better Responses	*Better responses* excelled in providing a clear and comprehensive explanation of how the standing waves are produced in the air column. Candidates effectively described how sound waves travel through vibrations in the pipe, mentioned the phase reversal of the wave upon reflection and explained that the superposition of incoming and reflected waves along the pipe leads to the formation of standing waves.
Image of Better Response	 Stationary waves are produced by superimposition of waves having same frequency and traveling in opposite direction. In air column waves reflect from one end and become out of phase 180° and when it direction is opposite to the coming wave it superimpose to create stationary waves.
Description of Weaker Responses	In *weaker responses*, some candidates gave answers that lacked proper support or were based on incorrect concepts. For instance, some candidates explained the production of standing waves in a string rather than in an air column. Others discussed constructive and destructive interference and the principle of superposition but did not specifically address standing waves in organ pipes.
Image of Weaker Response	 Stationary waves are produced in air coloumn gaps in a way that when two waves Y_1, Y_2 meet with each other at same time so the magnitude of resultant wave will be equal to the magnitude of both the wave

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

- Teachers are encouraged to incorporate visual diagrams and animations of the production of stationary waves in a sting and pipe that is open from both ends and closed from one end during their classroom teaching.
- Encourage candidates to make an experimental setup to demonstrate the formation of standing waves in their physics laboratory.
- Teachers are suggested to practice their students to pay attention to the wording of the question can help avoid misunderstandings and enhance accuracy in future assessments.
- Subject teachers should enhance their students' understanding of standing waves in both organ pipes and strings to better differentiate the formation of standing waves in various scenarios.

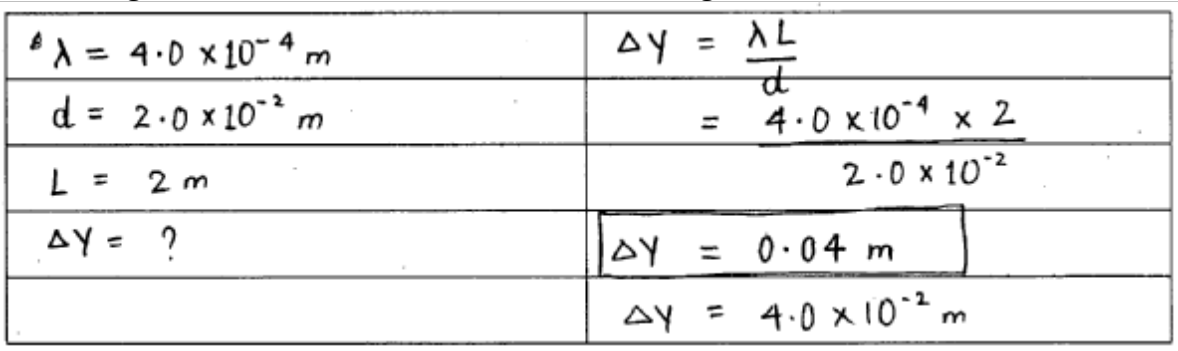
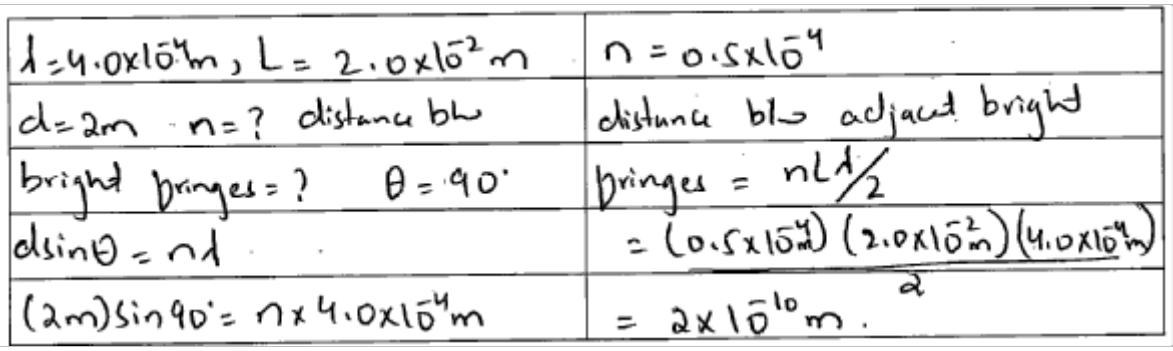
Question No. 8	
Question Text	The wavelength of light in the air is 4.0×10^{-4} m that is incident perpendicularly on a double slit and passes through the slits. The distance between the slits is 2.0×10^{-2} m and bright and dark fringes are observed on a screen 2 m away from the slits. Calculate the distance between the two adjacent bright fringes on the screen.
SLO No.	9.2.5
SLO Text	Derive relation for fringe spacing and use the relation in solving word problems.
Max Marks	03
Cognitive Level	A
Checking Hints	1 mark for writing the correct formula 1 mark for substituting the correct values 1 mark for writing the correct answer with SI unit
Overall Performance	The overall performance on this question was exceptionally well demonstrating a strong ability to identify the word problem as being related to Young's double-slit experiment. Their adept recognition of the context and specific experiment showcased a commendable understanding of the topic. Such well-attempted responses indicate a solid grasp of relevant concepts and principles.
Description of Better Responses	In *better responses*, candidates exhibited a good command over the subject, accurately extracting data from the question. They used the correct formula ($\Delta Y = \lambda L/d$) to calculate the answer and presented it in standard form. This accurate approach showcases their proficiency in handling calculations related to the interference of light.
Images of Better Responses	
Description of Weaker Responses	In *weaker responses*, candidates struggled to grasp the question's requirements, leading to errors in calculations and formula applications. Some even used inappropriate formulas and relations from Michelson interferometer, diffraction grating and Bragg's law.

Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

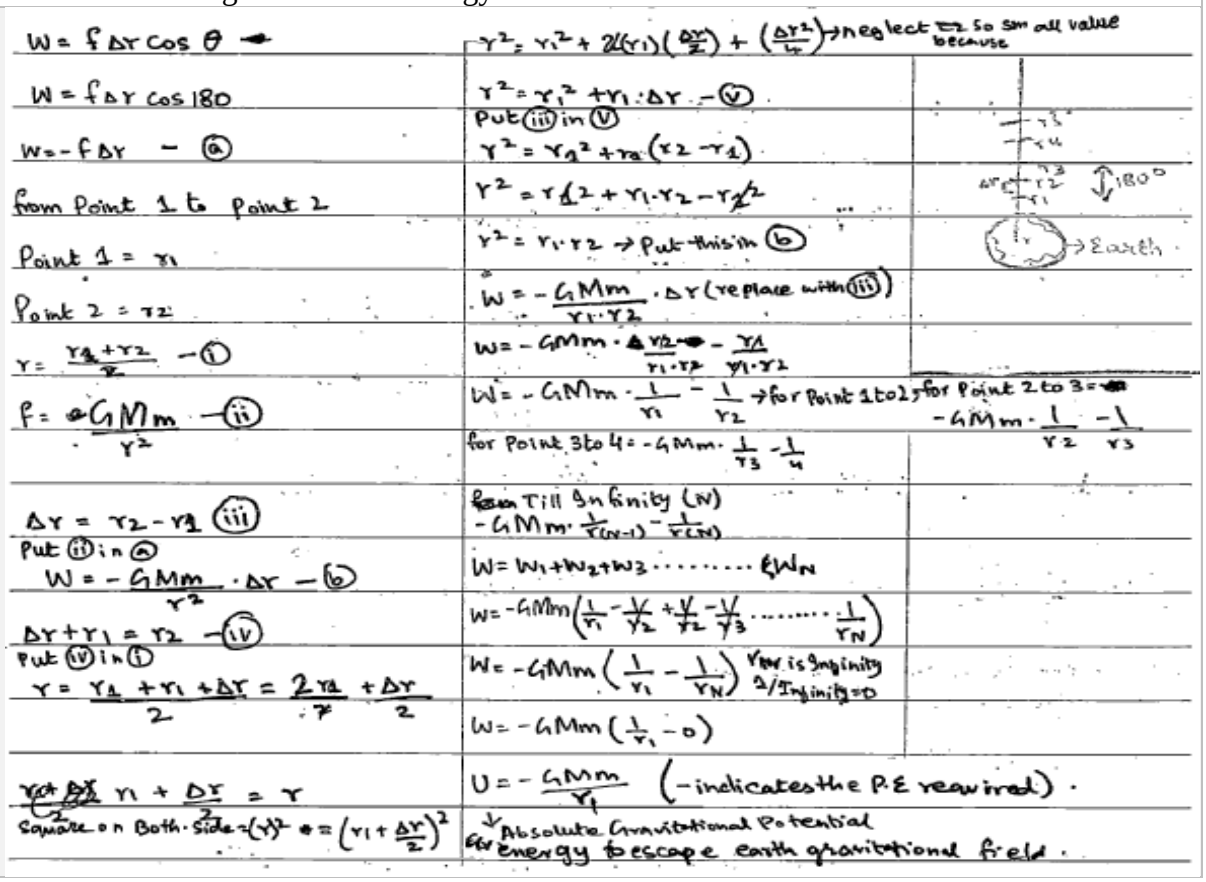
Maximising SLO Achievement	**Preferred Pedagogy Used for this SLO**	**Assessment Strategies**
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

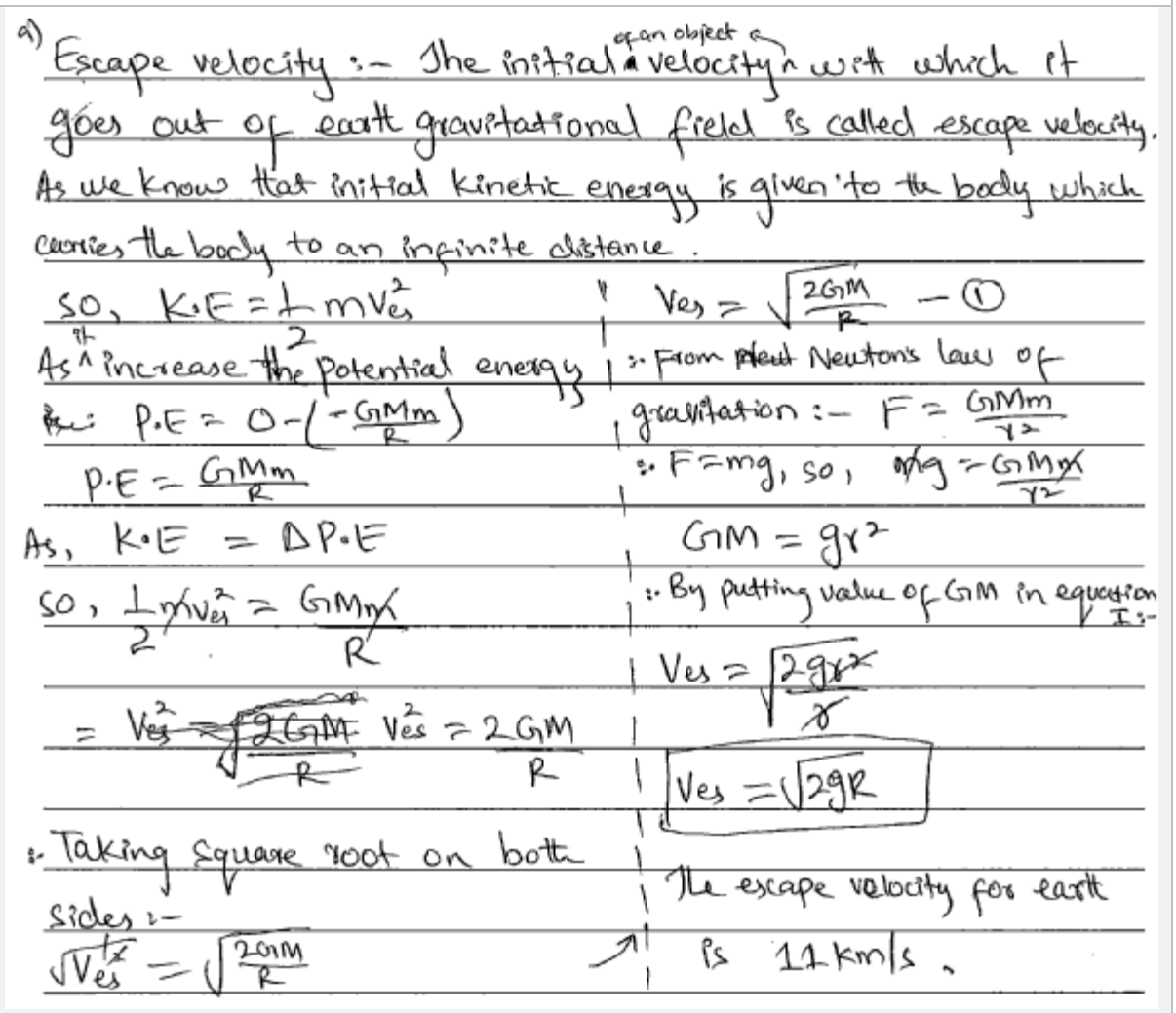
Any Additional Suggestion:

- Teachers are encouraged to create similar practice questions to help candidates improve their problem-solving skills and provide complete and accurate answers.
- Candidates are encouraged to thoroughly understand the question and identify the correct formulas for interference of light.
- Candidates should be guided by their teachers to provide appropriate application of relevant formulas which can support their development in this topic, resulting in more accurate responses in future assessments.

Extended Response Questions (ERQs)

Extended response questions offered a choice between parts 'a' and 'b'

Question No. 9a	
Question Text	Derive an expression for the energy required to escape a body from the Earth's gravitational field.
SLO No.	4.6.2
SLO Text	Derive an expression for absolute potential energy (PE).
Max Marks	07
Cognitive Level	U
Checking Hints	1 mark for each highlighted mathematical step (7 required)
Overall Performance	The majority of the candidates attempted this part of the question; however, their performance was below the desired level. Many candidates were distracted by the term 'escape' and derived the escape velocity formula instead of addressing the energy required to escape a body from Earth's gravitational field (absolute potential energy), as the question demanded. A few responses used irrelevant formulas and made incorrect statements. Candidates who correctly identified the question as relating to escape velocity provided the appropriate escape velocity formula, while those who accurately recognised the need for absolute energy formulated the correct relation. Overall, the responses indicate that while candidates generally have a good grasp of the content, misunderstandings of the question led to weaker performance.
Description of Better Responses	In *better responses*, candidates accurately identified the requirements of the question. They demonstrated a clear understanding of the concept by explaining that work done by a variable force requires dividing the path into small segments and calculating the work for each segment separately. Additionally, they correctly considered the upper point to be at infinity when determining the absolute energy.
Images of Better Responses	

Description of Weaker Responses	In *weaker responses*, most candidates were misled by the word 'escape' and focused on deriving escape velocity. Some responses involved calculating work done but were accompanied by unsupported statements.
Image of Weaker Response	

a) Escape velocity :- The initial velocity of an object with which it goes out of earth gravitational field is called escape velocity.

As we know that initial kinetic energy is given to the body which carries the body to an infinite distance.

So, $K.E = \frac{1}{2}mV_{es}^2$

As r increase the potential energy

$P.E = 0 - \left(-\frac{GMm}{R}\right)$

$P.E = \frac{GMm}{R}$

As, $K.E = \Delta P.E$

So, $\frac{1}{2}mV_{es}^2 = \frac{GMm}{R}$

$V_{es}^2 = \frac{2GM}{R}$

∴ Taking square root on both sides :-

$\sqrt{V_{es}^2} = \sqrt{\frac{2GM}{R}}$

$V_{es} = \sqrt{\frac{2GM}{R}} - (1)$

∴ From Newton's law of gravitation :- $F = \frac{GMm}{r^2}$

∴ $F = mg$, so, $mg = \frac{GMm}{r^2}$

$GM = gr^2$

∴ By putting value of GM in equation I :-

$V_{es} = \sqrt{\frac{2gr^2}{r}}$

$V_{es} = \sqrt{2gR}$

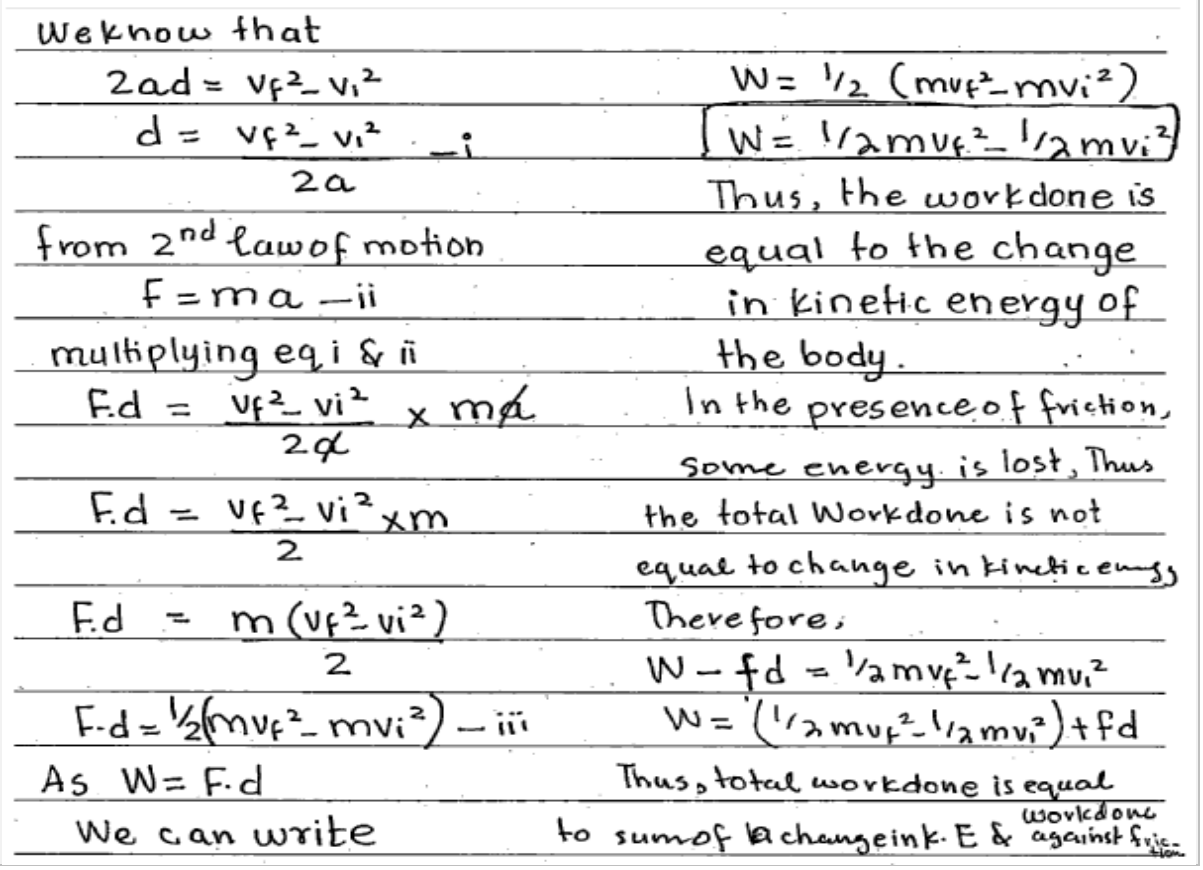
The escape velocity for earth is 11 km/s.

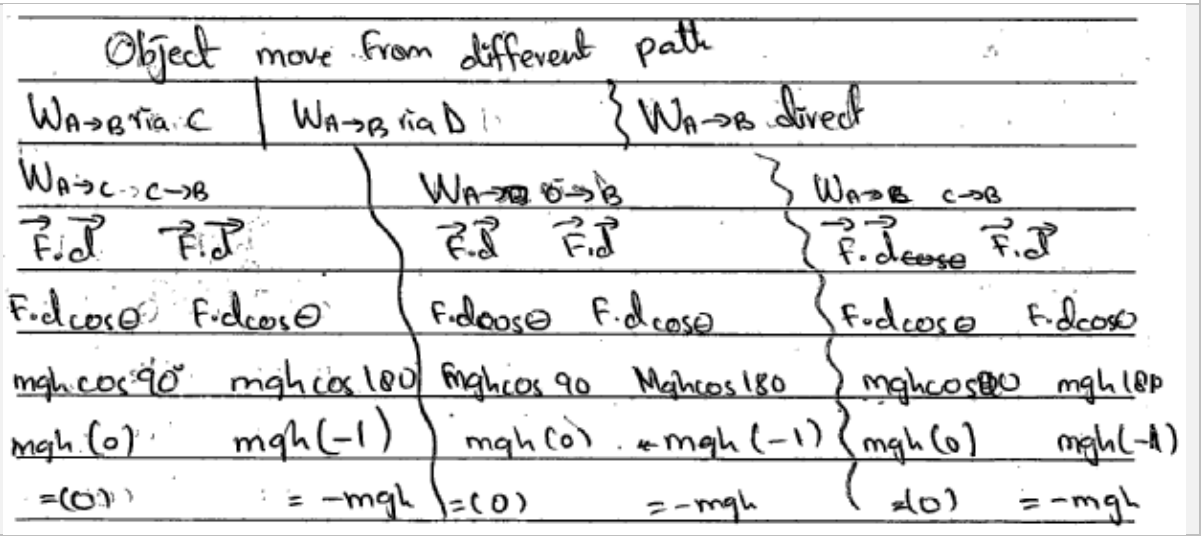
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

Encouraging candidates to closely examine the content asked in the question, recall their understanding made during classroom teaching and consider their distinguishing attributes can prevent such errors.

Question No. 9b	
Question Text	When a net force acts on an object, the work done on the object equals the change in its kinetic energy. Based on the given principle, derive work-energy equation in the presence of friction.
SLO No.	4.5.1
SLO Text	Deduce the relationship between energy and work a. when friction is present, b. when friction is not present.
Max Marks	07
Cognitive Level	U
Checking Hints	1 mark for each mathematical step (7 required)
Overall Performance	The overall performance of candidates on this question was generally good. Most candidates effectively derived the work-energy theorem in the presence of friction. However, some responses reflected a misunderstanding of the question, with candidates explaining the law of conservation of energy instead.
Description of Better Responses	In *better responses*, candidates demonstrated a solid understanding of the work-energy theorem. They provided a clear explanation of how frictional forces affect kinetic energy. By applying the equations of motion and Newton's second law of motion, these candidates effectively explained the given statement and established the work-energy equation in the presence of frictional forces.
Images of Better Responses	 We know that $2ad = v_f^2 - v_i^2$ $d = \frac{v_f^2 - v_i^2}{2a}$ – i from 2nd law of motion $F = ma$ – ii multiplying eq i & ii $F.d = \frac{v_f^2 - v_i^2}{2a} \times ma$ $F.d = \frac{v_f^2 - v_i^2}{2} \times m$ $F.d = \frac{m(v_f^2 - v_i^2)}{2}$ $F.d = \frac{1}{2}(mv_f^2 - mv_i^2)$ – iii As $W = F.d$ We can write $W = \frac{1}{2}(mv_f^2 - mv_i^2)$ $W = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$ Thus, the workdone is equal to the change in kinetic energy of the body. In the presence of friction, some energy is lost, Thus the total Workdone is not equal to change in kinetic energy. Therefore, $W - fd = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$ $W = (\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2) + fd$ Thus, total workdone is equal to sum of change in K.E & workdone against friction.

Description of Weaker Responses	In *weaker responses*, candidates provided unsupported answers and used irrelevant formulas to justify the given statement. Many of these responses incorrectly focused on explaining the law of conservation of energy or describing the gravitational field as conservative.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

Subject teachers must encourage their students to provide a mathematical explanation that demonstrates a clear understanding of how a statement can be justified using relevant formulas.

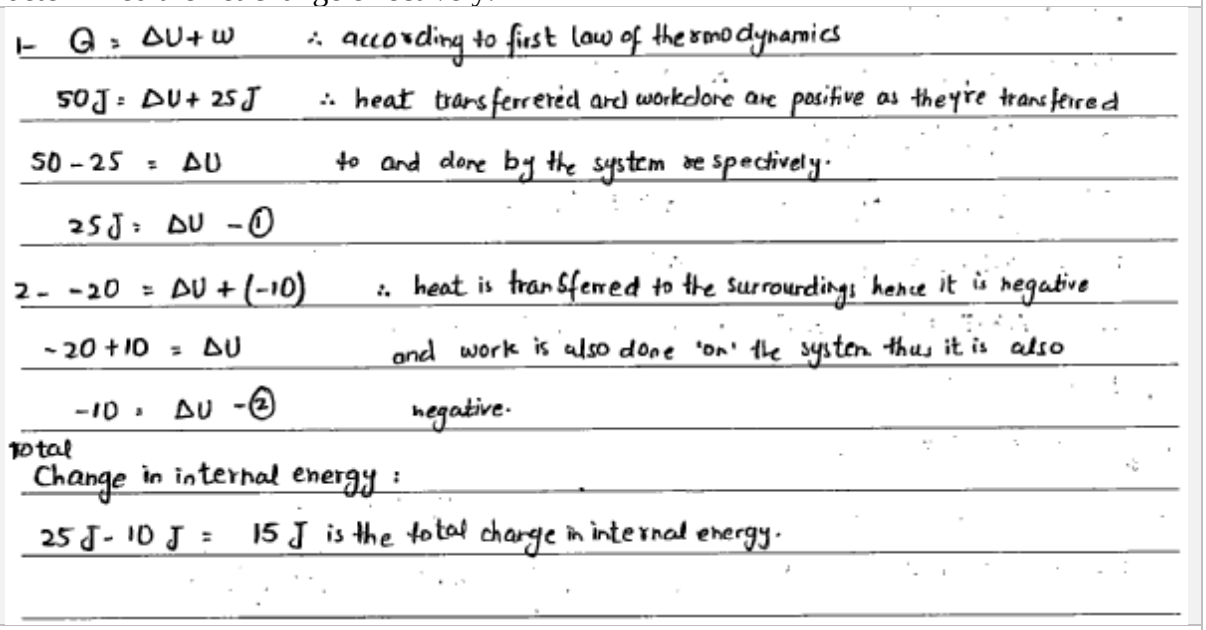
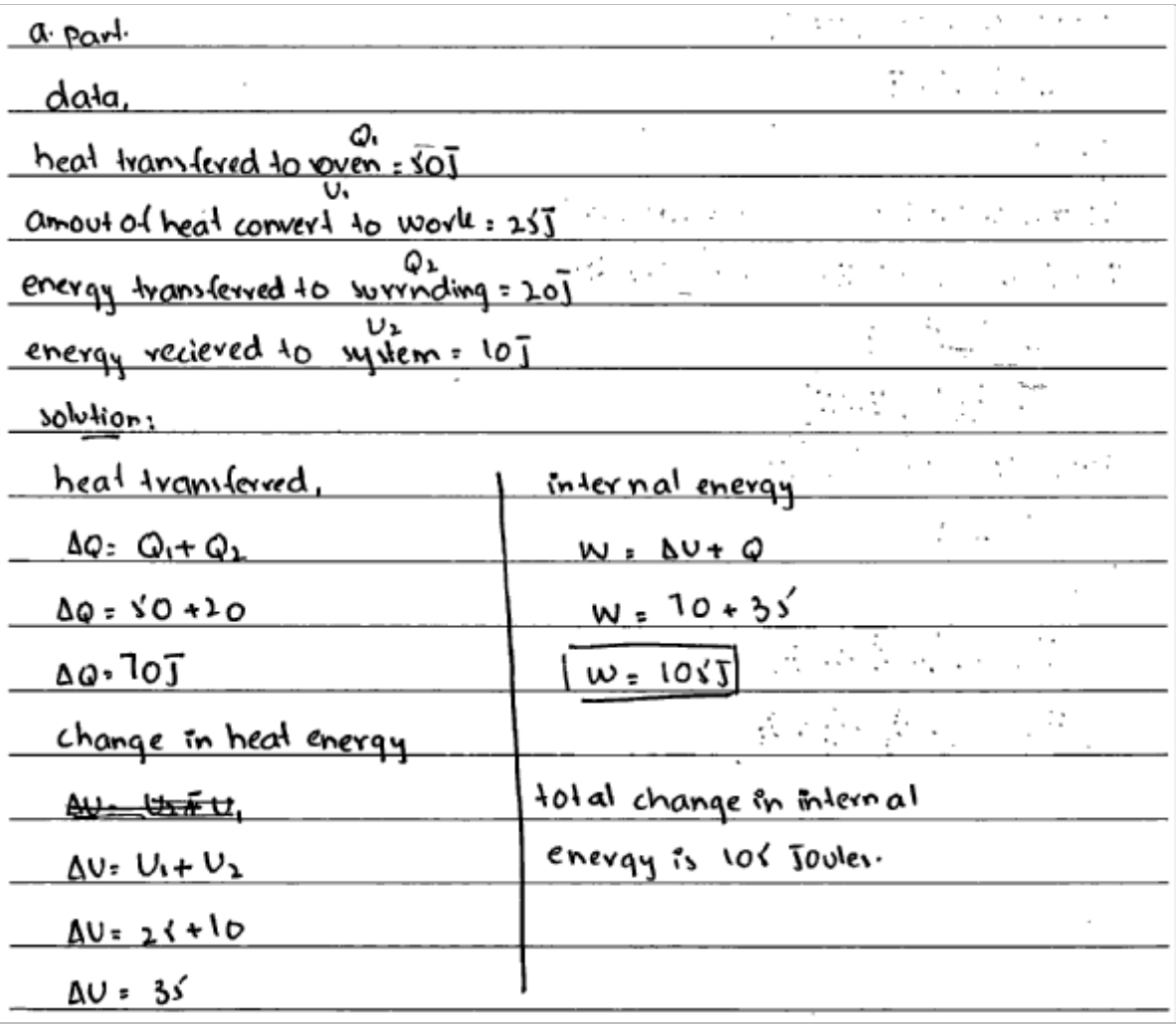
Question No. 10a	
Question Text	In a steel industry, 50 J of heat energy was transferred into the oven. The oven converted 25 J of heat energy into work done. At a later stage, the oven transferred 20 J of heat energy to its surrounding and received 10 J of workdone on the oven by its surrounding. Based on the given information, calculate the total change in internal energy of the oven.
SLO No.	10.4.3
SLO Text	Solve word problems related to the work done in thermodynamics system during a volume change.
Max Marks	07
Cognitive Level	A
Checking Hints	1 mark for writing each highlighted mathematical step (7 required)
Overall Performance	Overall, the majority of the candidates attempted this part of the question. Most responses clearly outlined the distinctions between heat energy and work done and correctly applied the sign conversion in the first law of thermodynamics. However, some responses contained errors in calculating the net change in internal energy.
Description of Better Responses	In *better responses*, candidates exhibited a good command over the subject, accurately extracting data from the question and using appropriate sign conversion for the heat input, heat output and work done on the system and by the system. They appropriately used the first law of thermodynamics to calculate the change in internal energy for each stage and determined the net change effectively.
Images of Better Responses	
Description of Weaker Responses	In *weaker responses*, candidates struggled to differentiate between heat energy input and output, as well as work done on and by the system. While some candidates made attempts, their efforts were often inaccurate. Additionally, some used incorrect formulas by placing the wrong position of variables at the wrong location.

Image of Weaker Response

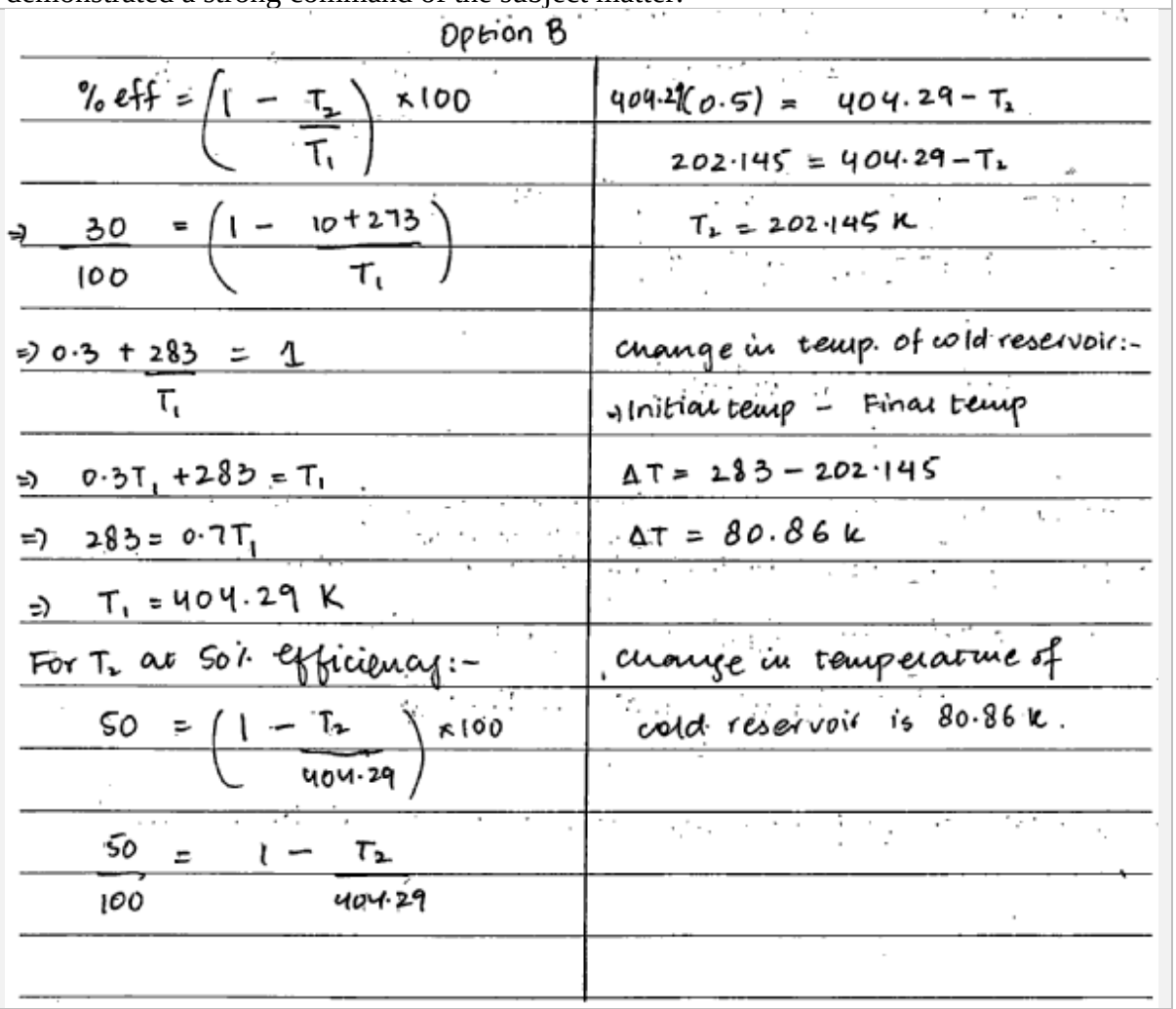


Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:

- Teachers are advised to develop similar practice questions to help candidates enhance their problem-solving skills and provide complete and accurate answers.
- Constructive feedback and targeted practice will guide candidates towards accurate differentiation, nurturing improved performance. Acknowledging their efforts while gently guiding them toward precision will enrich their overall learning journey.

Question No. 10b	
Question Text	A cold reservoir of a Carnot engine is at 10°C with an efficiency of 30%. To increase the efficiency of the engine to 50%, calculate the change in the temperature of the cold reservoir.
SLO No.	10.9.3
SLO Text	Derive the formula for efficiency of Carnot engine and use it in solving word problems\;
Max Marks	07
Cognitive Level	A
Checking Hints	1 mark for converting centigrade into kelvin 1 mark for each highlighted mathematical step (6 required)
Overall Performance	The overall performance of the cohort on this question was strong. Candidates showed a solid understanding by accurately identifying the given quantities and relating them to the second law of thermodynamics and the Carnot Engine. Their responses demonstrated a strong grasp of the relevant mathematical concepts.
Description of Better Responses	In *better responses*, candidates effectively extracted relevant data, identified the topic accurately and employed the correct formula along with appropriate substitutions, resulting in accurate outcomes presented in proper units. Their adept execution of calculations demonstrated a strong command of the subject matter.
Images of Better Responses	

Option B

$\%\ eff = \left(1 - \frac{T_2}{T_1}\right) \times 100$

$\Rightarrow \frac{30}{100} = \left(1 - \frac{10+273}{T_1}\right)$

$\Rightarrow 0.3 + \frac{283}{T_1} = 1$

$\Rightarrow 0.3T_1 + 283 = T_1$

$\Rightarrow 283 = 0.7T_1$

$\Rightarrow T_1 = 404.29\ K$

For T_2 at 50% efficiency:-

$50 = \left(1 - \frac{T_2}{404.29}\right) \times 100$

$\frac{50}{100} = 1 - \frac{T_2}{404.29}$

$404.29(0.5) = 404.29 - T_2$

$202.145 = 404.29 - T_2$

$T_2 = 202.145\ K$

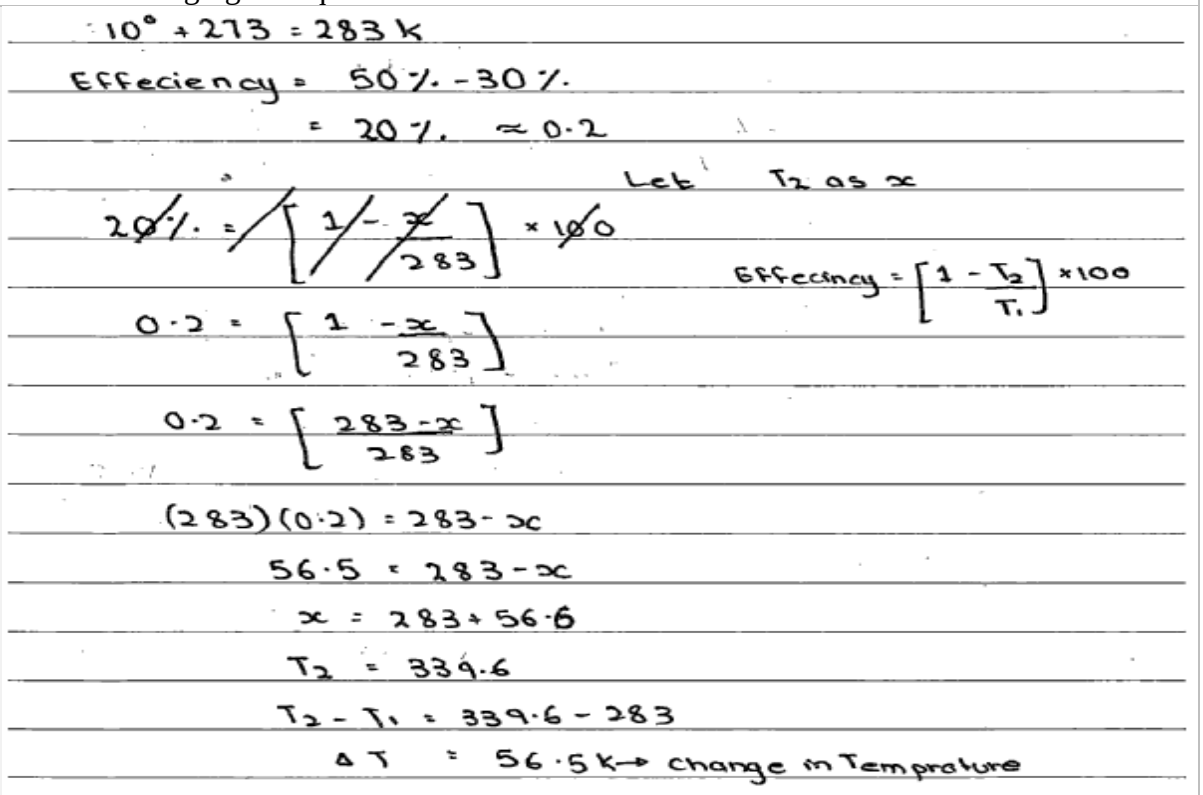
change in temp. of cold reservoir:-

→ Initial temp − Final temp

$\Delta T = 283 - 202.145$

$\Delta T = 80.86\ K$

change in temperature of cold reservoir is 80.86 K.

Description of Weaker Responses	In *weaker responses*, candidates provided inconsistent answers. Some correctly identified the concept and applied the appropriate formula but overlooked critical information provided in the question. Other candidates solved the problem but incorrectly interpreted the provided temperature as that of the hot reservoir. Additionally, some responses contained calculation errors in arranging the equations.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:

- Teachers are encouraged to create similar practice questions to help candidates improve their problem-solving skills and provide complete and accurate answers.
- Encouraging candidates to continue honing their problem-solving skills and ensuring precise application of formulas and units will further enhance their performance in similar scenarios.
- Regular practice of problems related to the efficiency of heat engines will enhance candidates' problem-solving skills, leading to more accurate responses in similar situations.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

Acknowledgements

The Aga Khan University Examination Board (AKU-EB) acknowledges with gratitude the invaluable contributions of all the dedicated individuals who have played a pivotal role in the development of the Physics HSSC-I E-Marking Notes.

We extend our sincere appreciation to Mr Kashif Hussain, Lead-Specialist in Physics at AKU-EB, for taking subject lead during the entire process of e-marking.

We particularly thank to Mr Saib Ahmed Khan, Lecturer, Govt. Degree Science and Commerce College, Orangi Township, Karachi, for evaluating each question's performances, delineating strengths and weaknesses in candidates' responses, and highlighting instructional approaches along with recommendations for better performance.

Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies along with suggestions for improvement.

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• Zain Muluk, Manager, Examination Development, AKU-EB

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• Ali Aslam Bijani, Manager, Teacher Support, AKU-EB

• Dr Naveed Yousuf, CEO, AKU-EB